

Customer Shopping Behavior Analysis Report

Executive Summary

This project aims to analyze customer shopping behavior and identify patterns that can help improve customer engagement, sales performance, and business decision-making. Using Python, SQL, and Power BI, customer transaction data was cleaned, analyzed, and visualized to generate actionable business insights.

Introduction

Understanding customer behavior is essential for retail businesses seeking to improve profitability and customer satisfaction. This project investigates purchasing patterns across customer demographics, product categories, payment methods, discounts, reviews, and shopping channels.

Business Question

How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

Dataset Overview

The dataset contains customer shopping information including:

- Customer Demographics
 - Product Categories
 - Purchase Amounts
 - Shopping Channels
 - Payment Methods
 - Discounts Applied
 - Customer Reviews and Ratings
 - Seasonal Purchase Information
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Methodology

Data Preparation

The dataset was cleaned using Python by:

- Removing duplicate records

- Handling missing values
- Correcting data types
- Preparing data for analysis

Exploratory Data Analysis

EDA was performed to understand:

- Customer distribution across demographics
- Popular product categories
- Spending behavior
- Seasonal trends
- Impact of discounts and reviews

SQL Analysis

SQL queries were used to:

- Segment customers
- Analyze purchase trends
- Identify high-performing categories
- Evaluate payment preferences
- Study customer loyalty patterns

Dashboard Development

An interactive Power BI dashboard was developed to visualize:

- Sales trends
- Customer demographics
- Product performance
- Payment method distribution
- Customer satisfaction indicators

Key Findings

Customer Segmentation

Customer purchasing behavior varies significantly across demographic groups. Certain customer segments consistently contribute higher sales and engagement.

Product Performance

Some product categories generate higher revenue and customer interest compared to others.

Discount Analysis

Discounts influence customer purchasing decisions and increase transaction volumes.

Shopping Channel Analysis

Online and offline channels demonstrate different purchasing patterns and customer preferences.

Payment Method Preferences

Digital payment methods are widely preferred among customers, indicating a shift toward cashless transactions.

Seasonal Trends

Customer purchasing activity fluctuates across seasons, highlighting opportunities for seasonal promotions and inventory planning.

Business Recommendations

Targeted Marketing

Focus marketing efforts on high-value customer segments identified during analysis.

Personalized Promotions

Offer personalized discounts and promotions based on customer preferences.

Inventory Optimization

Adjust inventory levels according to seasonal demand trends.

Customer Retention Programs

Implement loyalty programs to encourage repeat purchases and improve retention.

Product Strategy

Promote highly rated products and improve visibility for underperforming categories.

Conclusion

The analysis successfully identified key customer behavior patterns and business opportunities. The insights generated can help management make informed decisions regarding marketing strategies,

customer engagement, inventory management, and product planning. By leveraging data-driven decision-making, the company can improve customer satisfaction and drive long-term business growth.

Tools Used

- Python
- Pandas
- NumPy
- SQL
- Power BI
- Jupyter Notebook

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